Astik Srivastava

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Education

- **International Institute of Information Technology** — Hyderabad, India
  Master of Science (Research) - Computer Science and Engineering; GPA: 9.17/10 — *Jan 2025 - Jan 2027*

  Courses: *Computer Vision, Robotics: Planning and Navigation, Mobile Robotics, Statistical methods in AI*

- **Delhi Technological University** — New Delhi, India
  Bachelor of Technology - Engineering Physics; GPA: 7.99/10 — *Dec 2020 - Aug 2024*

  Courses: *Computer Vision, Deep learning, Machine learning, Signals and systems, Microprocessors and interfacing, Digital design, Mathematical Physics, Computational methods, Engineering Mechanics, classical and quantum physics*

Skills Summary

- **Languages**: Python, C++, C, Rust
- **Softwares**: MATLAB, Git, Docker, Solidworks, Ansys
- **Platforms**: Linux, Windows, ROS1/ROS2
- **Hardwares**: Arduino, Raspberry pi, Nvidia Jetson, Pixhawk

Experience

- **Google Summer of Code** — Offsite
  Open Source contributor (Part-time) — *May 2026 - August 2026*
  - **Kornia-SLAM: Robust visual-inertial SLAM**: Development of a vision based simultaneous localization and mapping algorithm written in Rust for allowing robots to localize themselves in indoor and outdoor environments using low compute hardware.
- **Google Summer of Code** — Offsite
  Open Source contributor (Part-time) — *May 2024 - August 2024*
  - **All-in-One System Identification Toolkit for ArduPilot Platforms**: Development of a system identification toolbox for UAVs, which can extract dynamic model parameters from flight data.
- **Addverb Technologies** — Onsite
  Robotics Intern (part-time) — *September 2023 - February 2024*
  - **Simulation development for warehouse automation robots**: Working on simulation development to design, test and validate motion planning and perception algorithms with applications in mobile robotics
- **EndureAir systems** — Onsite
  Embedded Software intern (part-time) — *June 2023 - August 2023*
  - **Embedded software development for unmanned aerial systems**: Working on embedded applications that will work in conjugation with the avionics stack of multiple unmanned aerial systems.

Publications

- **Journal paper: MORSE: Moving-Object Segmentation and Metric Localization via Radar-Visual Bundle Adjustment**: Submitted for IEEE-RAL Tech: C++, OpenCV, ONYX, Python, (October '26)
- **Conference paper: SPOT: Spatio-Temporal Obstacle-free Trajectory Planning for UAVs in an Unknown Dynamic Environments**: Accepted for ICRA 2026 Tech: C++, OpenCV, ONYX, Python, (September '25) See paper
- **Conference paper: Learning Real-time Glass Segmentation using Radar-Depth Fusion**: Accepted for IROS 2026 Tech: Pytorch, OpenCV, ONYX, Python, (March '26) See paper
- **Conference paper: S3: Stratified Sparse Stereo for Deterministic Onboard Obstacle Avoidance on Resource-Constrained Nano-UAVs**: Submitted for ICRA 2027 Tech: C++, OpenCV, ONYX, Python, (September '26)
- **Conference paper: A Modified Artificial Potential Field for UAV Collision Avoidance**: Accepted for ICUAS 2023. Tech: MATLAB, Python(November 22) See paper
- **Conference paper: Depth Estimation Using Monocular Camera for Real-World Multi-Object Grasp Detection for Robotic Arm**: Presented at ICSCC 2023. Tech: MATLAB, Python, Computer Vision(September 23) See paper

Projects

- **Radar-based dynamic object perception**: Image frame dynamic object segmentation using radar-rgb fusion on nuscenes dataset and simultaneous ego-velocity estimation. Accepted in IEEE IROS 2026 LTP Workshop. Tech: PyTorch, Python, C++. (June 26)
- **Motion Planning for 8-Joint micro-Quadruped**: Development of an MPPI based motion planner for enabling collision avoidance for 8 joint micro quadruped robot. Tech: Python, Pybullet (Sept 26)
- **Embedded LQR attitude control of a quadcopter UAV**: Development of a linear quadratic regulator based attitude control for a quadcopter. Controller was integrated with ardupilot's custom control block. Tech: Python, C++, Matlab(Dec 23)

Position of Responsibility

- **Team Captain: Unmanned Aerial Systems-DTU (August 2022-August 2023)**: Directly responsible for mentoring and managing a robotics research team of 20 undergraduate students.

Honors and Awards

- 3rd position in the International Micro Aerial Vehicle Competition (outdoor) 2023.
- All India 3rd rank in SAE Drone Development Challenge 2023 (Regular Category)